Resume

ikhan1@unb.ca **Irtiza Masrur Khan** +1 (506) 4282899

LinkedIn: www.linkedin.com/in/irtiza-khan-512992193

I am a Canadian graduate researcher in fluid dynamics, focused on integrating Machine Learning with Computational Methods and Design Optimization. I am currently developing workflows that combine automated CAD modeling, meshing, CFD, and Surrogate-Assisted Optimization to accelerate performance prediction and shape optimization (marine propellers, airfoil slats).

Education

MSc, Mechanical Engineering — University of New Brunswick, Fredericton, NB, Canada
May 2024 – Present • **CGPA: 4.3/4.3**

BSc, Mechanical Engineering — University of New Brunswick, Fredericton, NB, Canada
Sept 2018 – May 2024 • **CGPA: 3.9/4.3 (Dean's List)**

Technical Skills

- **Programming**: Python, MATLAB, C++
- **Modeling & Meshing**: PythonOCC/OpenCASCADE, Autodesk Fusion 360, Pointwise
- **Simulation & Analysis**: ANSYS CFD (Fluent & CFX for steady & unsteady flows)
- **AI/ML & Optimization**: Machine learning libraries (e.g., scikit-learn, GPflow), multi-objective & metaheuristic evolutionary algorithms, design of experiments (DOE) frameworks

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Research Experience

Design & Optimization of Marine Propellers and Slats — Graduate Researcher, UNB
May 2024 – Present (Full-Time)

- Developed Python scripts using PythonOCC/OpenCASCADE (OCCT) kernel to automate 3D blade geometry generation, enabling rapid iteration in optimization loops.
- Designed and implemented end-to-end workflows integrating geometry parametrization, automated meshing, CFD evaluation (via ANSYS), and optimization frameworks.

- Deeply investigated multiple evolutionary algorithms (*e.g.,* Genetic algorithm, Particle Swarm Optimization, Differential Evolution, NSGA-II, *etc.*) to select the best-fit algorithm for an efficient optimization workflow.
- Worked on surrogate model development (e.g., Gaussian Process with infill strategies) to enable high-fidelity simulations (CFD) based shape optimization.
- Applying the same framework to a multi-component (thick airfoils with a slat) optimization and analyzing performance comparison with a Particle Swarm Optimization based workflow.
- Exploring different Gaussian Process variations, infill strategies, and other machine learning methods to enhance optimization efficiency further.

Geometry Parametrization Strategies for Marine Propellers — Undergrad Researcher, UNB

May 2023 – May 2024 (Full-time & Part-time)

- Developed code that automatically generates a 3D point cloud of a propeller blade based on standard propeller design parameters (pitch distribution, chord distribution, skew distribution, etc.).
- Performed turbomachinery simulations in ANSYS CFX: meshing, setting up simulation parameters & boundary conditions, and post-processing.

Research Interests

- **Fluid Dynamics & High-fidelity CFD**: Steady & unsteady flows, turbulence, propulsors, hydrodynamics, and aerodynamics.
- **AI for Engineering:** Surrogate models (*e.g.,* Gaussian Processes, Radial Basis Functions), scientific machine learning, reduced-order modeling, deep neural networks, and reinforcement learning.
- **Design & Optimization:** Multi-objective algorithms (e.g., NSGA-II), evolutionary strategies, algorithm development, automated CAD & mesh generation, and high-fidelity shape optimization.
- **Applied Mathematics:** Practical applications of numerical methods for complex engineering problems.

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Publications

1. **Khan, I. M.**, *et al.* "Automated 3D Propeller Modeling for Integration into a Shape Optimization Workflow." Proc. CSME/CFDSC/CSR 2025, Montréal, Canada.
2. *(Under review)* **Khan, I. M.**, *et al.* "Surrogate-Assisted Propeller Shape Optimization in Hull Wake Using 3D RANS Simulations." *Abstract accepted by the Symposium on Marine Propulsors (SMP) 2026*, St. John's, NL, Canada.
3. *(Under review)* **Khan, I. M.**, *et al.* "Sample-Efficient Slat Optimization for Thick Airfoils: GP-BO Compared with PSO." *Extended Abstract Submitted to AIAA Aviation Forum 2026*, San Diego, CA, USA.

Teaching Experience

Graduate Teaching Assistant, UNB

September 2024 – Present (Part-Time)

- **Fluid Mechanics:** Led tutorials and office hours, marked exams and assignments, and supported students with core concepts.
- **Fluid Mechanics Laboratory:** Ran lab sections for large groups, assessed reports, and provided pre-lab briefings and in-lab guidance.
- **Fluid Systems and Design:** Organized and supervised field experiments and evaluated technical reports.
- **Thermodynamics:** Delivered weekly tutorials to a large class, assisted with exam and assignment marking.

Other Work Experiences

Direct Support Specialist, CBI Health Group

April 2019 – Present (Part-Time)

- Trained in providing direct support and care to people with disabilities (*e.g.,* autism spectrum disorder, neurological development issues, physical disabilities).
- Trained in non-violent crisis intervention (NVCI) and certified first-aid provider.
- Involved in supporting clients with their day-to-day necessities while actively working towards meeting long-term goals in collaboration with behavioral and medical specialists.

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Co-founder & Chief Operating Officer, Tutor on Time Tech. Ltd.

January 2021 – December 2023 (Full-Time Co-op & Part-time)

- Co-founded an online teaching platform for post-secondary students with support from UNB's Technology Management & Entrepreneurship (TME) program.
- Raised funds through multiple business pitch competitions and funding opportunities, and maintained communications with multiple consumer institutions: universities and colleges in New Brunswick.
- Managed a small team of software engineers and prototyped an online classroom system like MS Teams/ Zoom, designed to maximize learning experience.

Awards & Scholarships

- Accelerated Master's Scholarship (2024)
- UNB Graduate Scholarship (2024)
- H. E. McKeen Scholarship (2021, 2022)
- George Cedric Ferguson Memorial Engineering Bursary (2022)
- Edwin Jacob Special University Scholarship (2019, 2021)
- Sydney Acker Scholarship (2020)
- UNB Engineering Alumni Scholarship (2019)
- UNB Scholarship for Academic Excellence (2018)

Personal Interest & Activities

- **Bookworm:** Reading science fiction and non-fiction books, especially on Black Holes and Genetic Engineering.
- **Novice Chess Player:** Reading chess theory and improving in the game.
- **Avid Hiker:** Hiking on the beautiful nature trails of NB, camping, and fishing.
- **Beginner Rock Climber:** Learning the basics with hopes of doing non-competitive, recreational bouldering and rope-climbing.